17–RUDDER THEORY

17.1 Rudder

The rudder is the primary means for controlling the direction of motion of a ship.

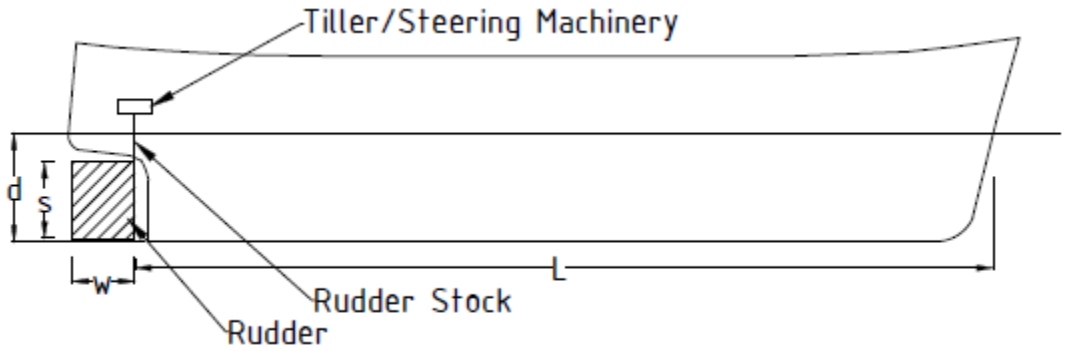


Figure 17.1 – Rudder & Ship

The profile or shape of a rudder is described by its:

Tip chord (c_R):	The width at the top
Root chord (c_T):	The width at the bottom
Mean chord (c):	$(c_R + c_T)/2$
Taper ratio (TR):	c_T/c_R
Span (s):	Its depth
Area (A):	s x c
Aspect ratio (AR):	$s/c = s^2/A \approx 2$

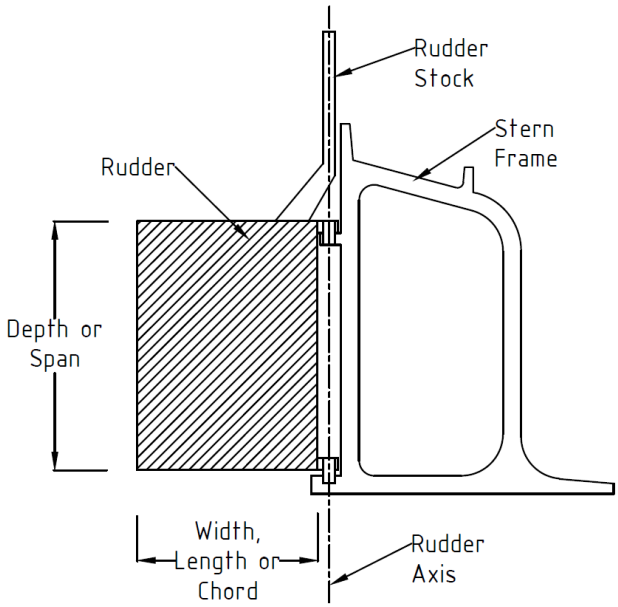


FIGURE 17.2 – Rudder

The area of a rudder suitable for a particular vessel is obtained by empirical formulae for which there are a number of examples developed over the years from

experience. Values usually lie between 1.4 to 1.6% of the projected area of the ship below the load waterline. Typical examples are:

Example 1:

$$A = K \times L \times d$$

Example 2 (From DNV):

$$A = \frac{L \times d}{100}\left(1 + 25\left(\frac{B}{L}\right)^2\right)$$

Where: K = Coefficient
= 1.2–1.7 for container & passenger ships
= 1.5 for general cargo ships
= 1.7 for oil tankers & bulk carriers
L = LBP
d = Load draught
B = Breadth at midship

The height of rudder to its width is defined by the:

$$Aspect\ Ratio = \frac{s}{w}$$

This is usually about 2.

17.2 Rudder Types

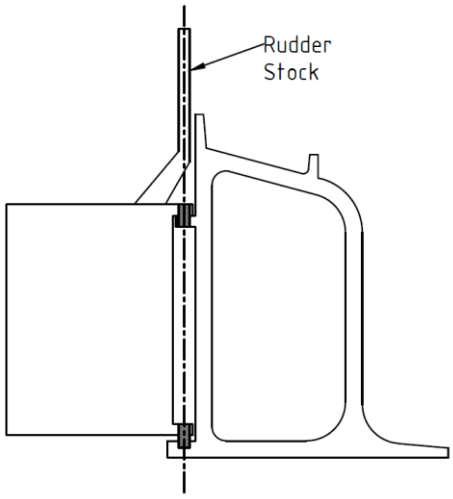


FIGURE 17.3 – Unbalanced Rudder

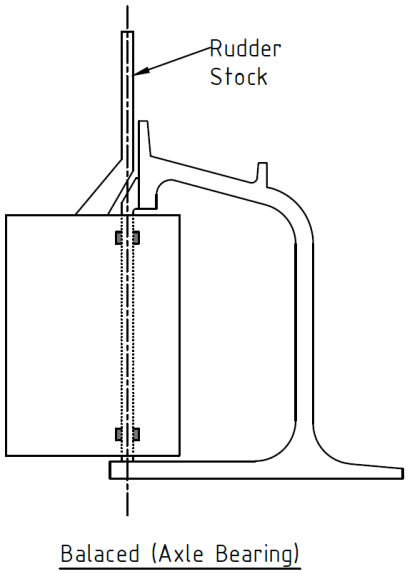


FIGURE 17.4 – Balanced Rudder

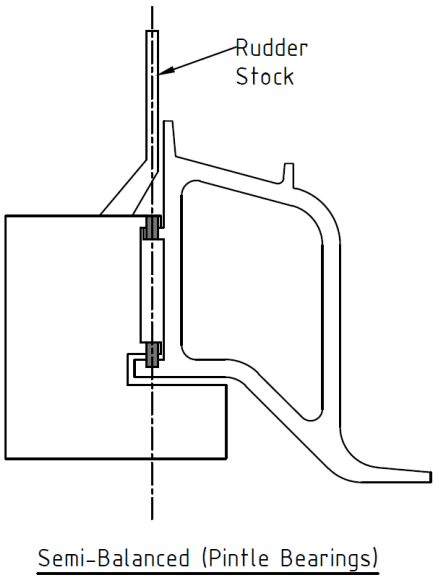


FIGURE 17.5 – Semi-Balanced Rudder

17.3 Force on Rudder

When a rudder is moved through water at zero angle of incidence (i.e. in line with the incoming flow) no transverse force will be generated and the ship will remain on a straight course.

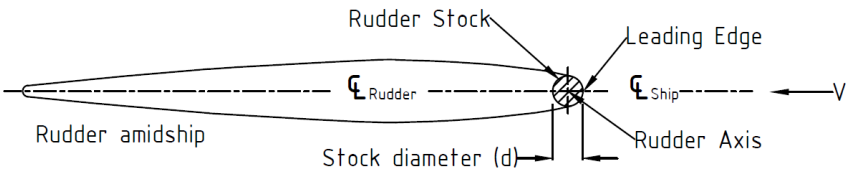


Figure 17.6 – Rudder Amidship

When now the rudder is turned to an angle of incidence or attack (Alpha) to the centerline of ship, the incoming flow hits the rudder at the centre of pressure, a distance (b) from the leading edge, and generates lift (L) and drag (D) forces as shown in Figure 17.7.

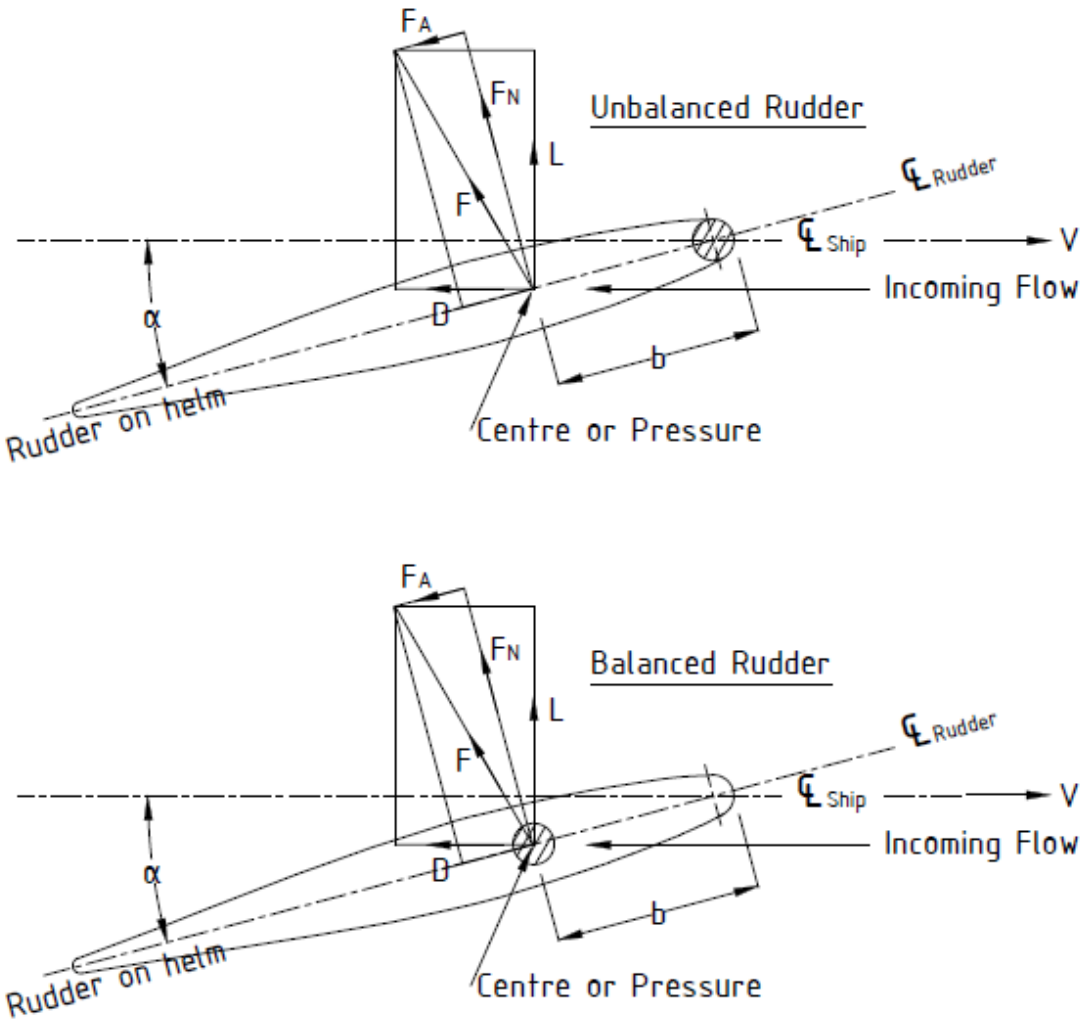


Figure 17.7 – Rudder on Helm

The resultant of these two is a hydrodynamic force (F) which is determined by the expression:

$$F = kAV^2$$

Where:

- k is a coefficient which is dependent on:
  - Cross-sectional shape of rudder
  - Angle of incidence
  - Density of water
  - Aspect ratio
- A is area of rudder in (m^2)
- V is speed of incoming flow or ship speed in (m/s). However, due to the effect of propeller race on the water this velocity may be increased by a factor

of 1.3 for a rudder behind the propeller & 1.2 for a centerline rudder behind twin screws.

The distance (b) varies with alpha and may be determined by a formula developed by Joessel as follows:

$$b = w(0.195 + 0.305sin\alpha)$$

The resultant force (F) can be further resolved into components along and normal to the axis of rudder. The axial component (F_A) will act in opposition to motion and slow down the ship while the normal component (F_N) will act perpendicularly to the rudder axis and cause a transverse push on the stern. It is hence responsible for the turning of the ship and referred to as the Rudder Force.

$$F_N = kAV^2sin\alpha$$

The relationship between lift (L) and (Alpha) increases linearly as shown in Figure 17.8 until between 35 & 45 degrees stalling occurs. The maximum rudder angle is hence limited to 35 degrees on either side.

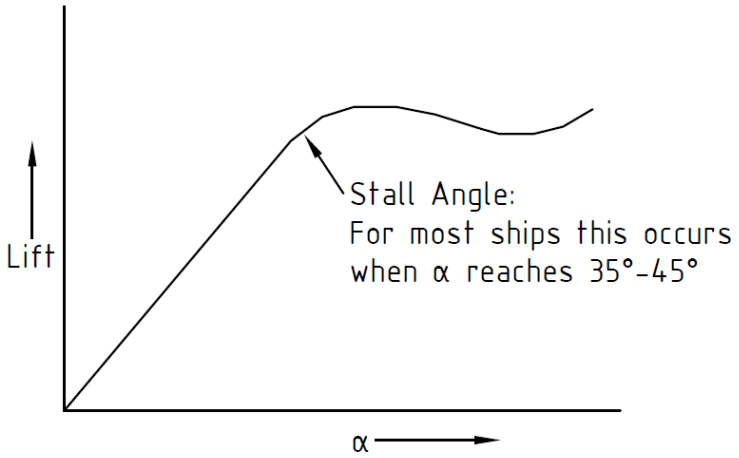


Figure 17.8 – Lift/Angle of Attack Curve

17.4 Torque on Rudder

In the process of turning rudder force F_N causes a turning moment (M) to be applied where:

$$M = F_N \times b$$

This will be equally resisted by the rudder stock. The torque (T) on the stock will therefore be:

$$T = F_N \times b$$

Now from the Basic Torsion Equation:

$$\frac{T}{J} = \frac{q}{r}$$

Where:

- r is radius of stock's cross-section (m)
- J is the second polar moment of area (m^4), and equals:

$$\frac{\pi r^4}{2}, \text{ or } \frac{\eta d^4}{32}$$

- q is the maximum allowable stress (N/m^2)

By means of this equation therefore, the strength and sizing of a rudder stock is determined.

For a balanced rudder there is a particular angle at which the centre of pressure coincides with the centre of stock which, in which condition F_N imposes no stress on the rudder stock.

17.5 Angle of heel when turning

The action of the rudder force when turning a ship is analyzed as follows:

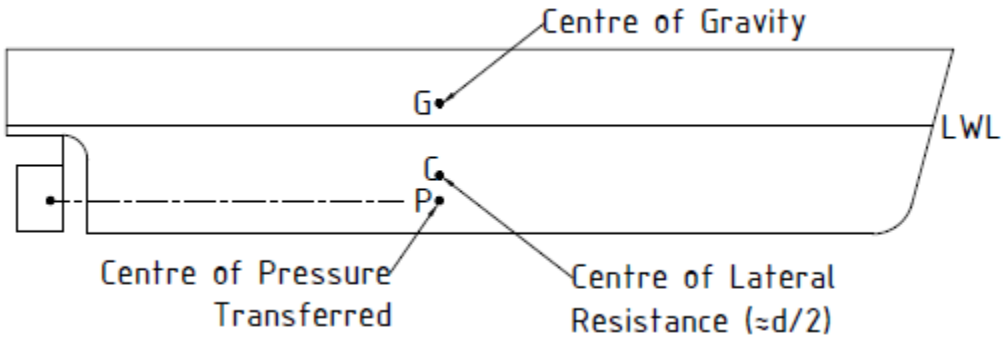


Figure 17.9 – Definition of Various Centres

In Figure 17.9, (G) is the position of the vertical centre of gravity (KG). Let (P) be the vertical position of the Centre of Pressure when transferred to a longitudinal axis through G, and let (C) be the centroid of the projected underwater area. It is referred to as the Centre of Lateral Resistance and its height is considered to be approximately equal to half draught.

Initial Angle of Heel

During the first stage of turn when helm is put on the rudder, force F_N comes into action as in Figure 17.10(a). By replacing F_N with a system of forces as in Figure (b), it is seen that a couple (F_N.x) comes into action to turn the ship while an unbalanced transverse force F_T begins to sway the ship from its path against water resistance which acts through C as in Figure (c). However, due to rotational inertia, turning does not immediately take effect, rather F_T heels the ship inwards about (C) towards the centre of the turning circle as in Figure (d).

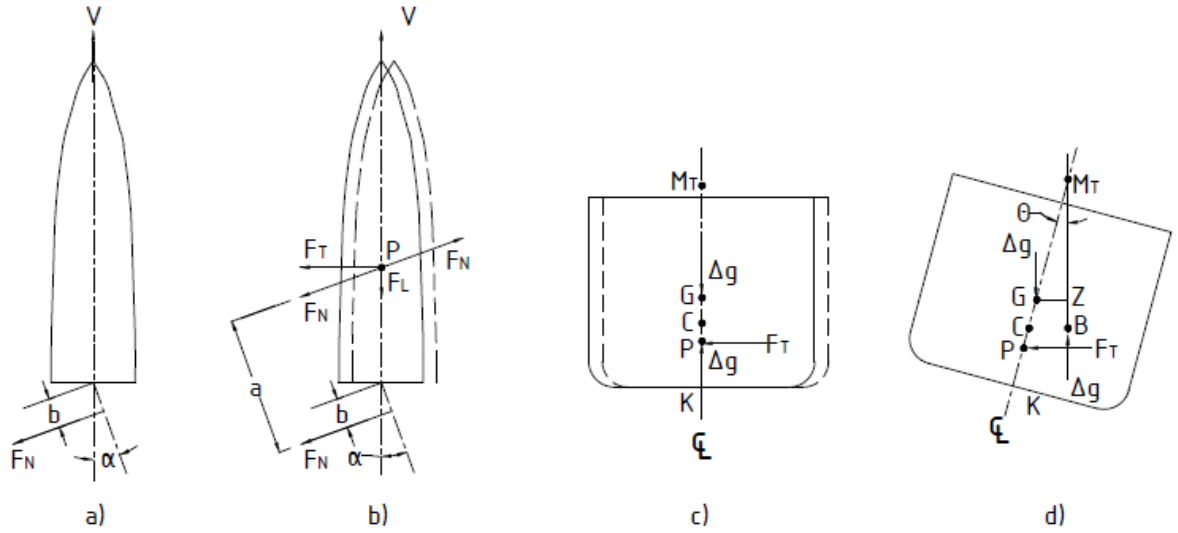


Figure 17.10 – Initial Angle of Heel

A counter displacement moment then comes into action until at an angle of balance when the rotational inertia has been overcome:

$$HM = RM$$

$$F_T \times PC cos\theta = \Delta g \times GZ = \Delta g \times GM_T sin\theta$$

From which the initial angle of heel is given by the expression:

$$tan\theta = \frac{F_T . PC}{\Delta g . GM_T}$$

Steady Angle of Heel

With rotational inertia overcome, the ship now begins to turn and a centrifugal force comes into action. Acting like a giant hydrofoil, lift and drag forces are also generated which can then be combined into a resultant force (R) as in Figure 17.11(e).

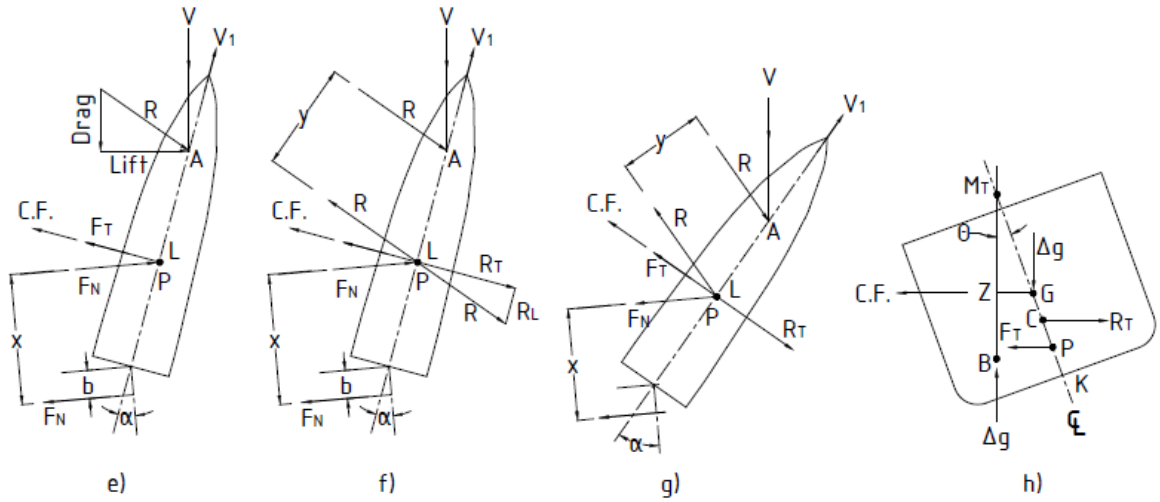


Figure 17.11 – Steady Angle of Heel

Again (R) may be represented by a system of force R_T and a couple (R.y) as in Figure (f). The combination of couples (F_N.x) and (R.y) increases the turn as in Figure (g), while vertical moment (R_T.G) overcomes moment (F_T.G) by canceling it out. Under the influence of the centrifugal force, the ship begins to heel away from the centre of the turning circle. This brings into action an opposing displacement moment which at an angle of balance will make:

$$HM = RM$$

$$C.F. \times CG cos\theta = \Delta g \times GZ = \Delta g . GM_T sin\theta$$

Now:

$$C.F. = \frac{\Delta V^2}{r}$$

Therefore:

$$\frac{\Delta V^2}{r} \times CG cos\theta = \Delta g . GM_T sin\theta$$

And:

$$tan\theta = \frac{V^2 CG}{g . r . GM_T}$$

17.6 Geometry of turning circle

Actions of the rudder as described above results in a path of vessel as shown in Figure 17.12

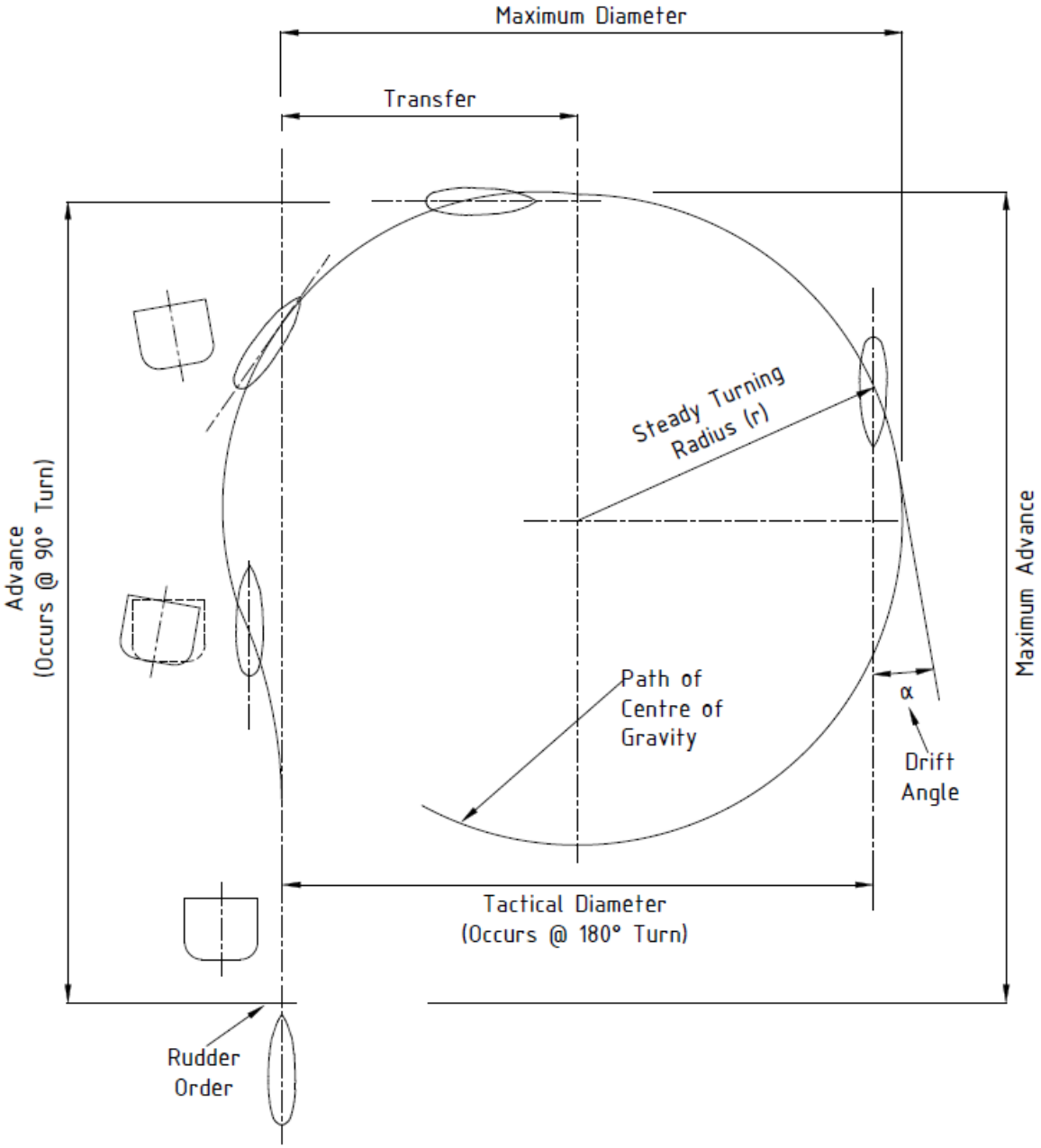


Figure 17.12 – Geometry of Turning Circle